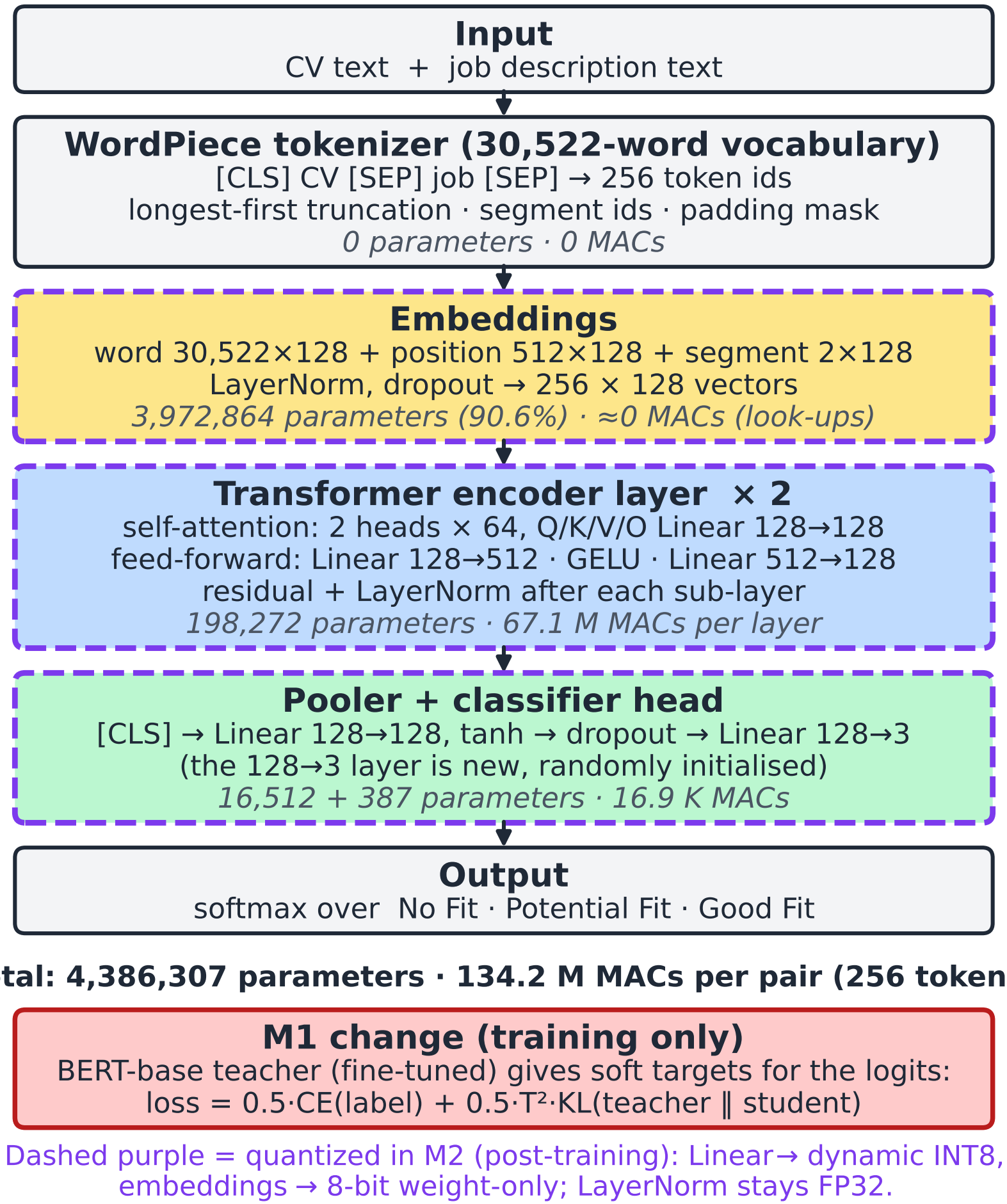
M0: BERT-Tiny cross-encoder (L=2, H=128, A=2)
Input
CV text + job description text
WordPiece tokenizer (30,522-word vocabulary)
[CLS] CV [SEP] job [SEP] → 256 token ids
longest-first truncation · segment ids · padding mask
0 parameters · 0 MACs
Embeddings
word 30,522×128 + position 512×128 + segment 2×128
LayerNorm, dropout → 256 × 128 vectors
3,972,864 parameters (90.6%) · ≈0 MACs (look-ups)
Transformer encoder layer × 2
self-attention: 2 heads × 64, Q/K/V/O Linear 128→128
feed-forward: Linear 128→512 · GELU · Linear 512→128
residual + LayerNorm after each sub-layer
198,272 parameters · 67.1 M MACs per layer
Pooler + classifier head
[CLS] → Linear 128→128, tanh → dropout → Linear 128→3
(the 128→3 layer is new, randomly initialised)
16,512 + 387 parameters · 16.9 K MACs
Output
softmax over No Fit · Potential Fit · Good Fit
Total: 4,386,307 parameters · 134.2 M MACs per pair (256 tokens)
M1 change (training only)
BERT-base teacher (fine-tuned) gives soft targets for the logits:
loss = 0.5·CE(label) + 0.5·T²·KL(teacher ‖ student)
Dashed purple = quantized in M2 (post-training): Linear→ dynamic INT8,
embeddings → 8-bit weight-only; LayerNorm stays FP32.